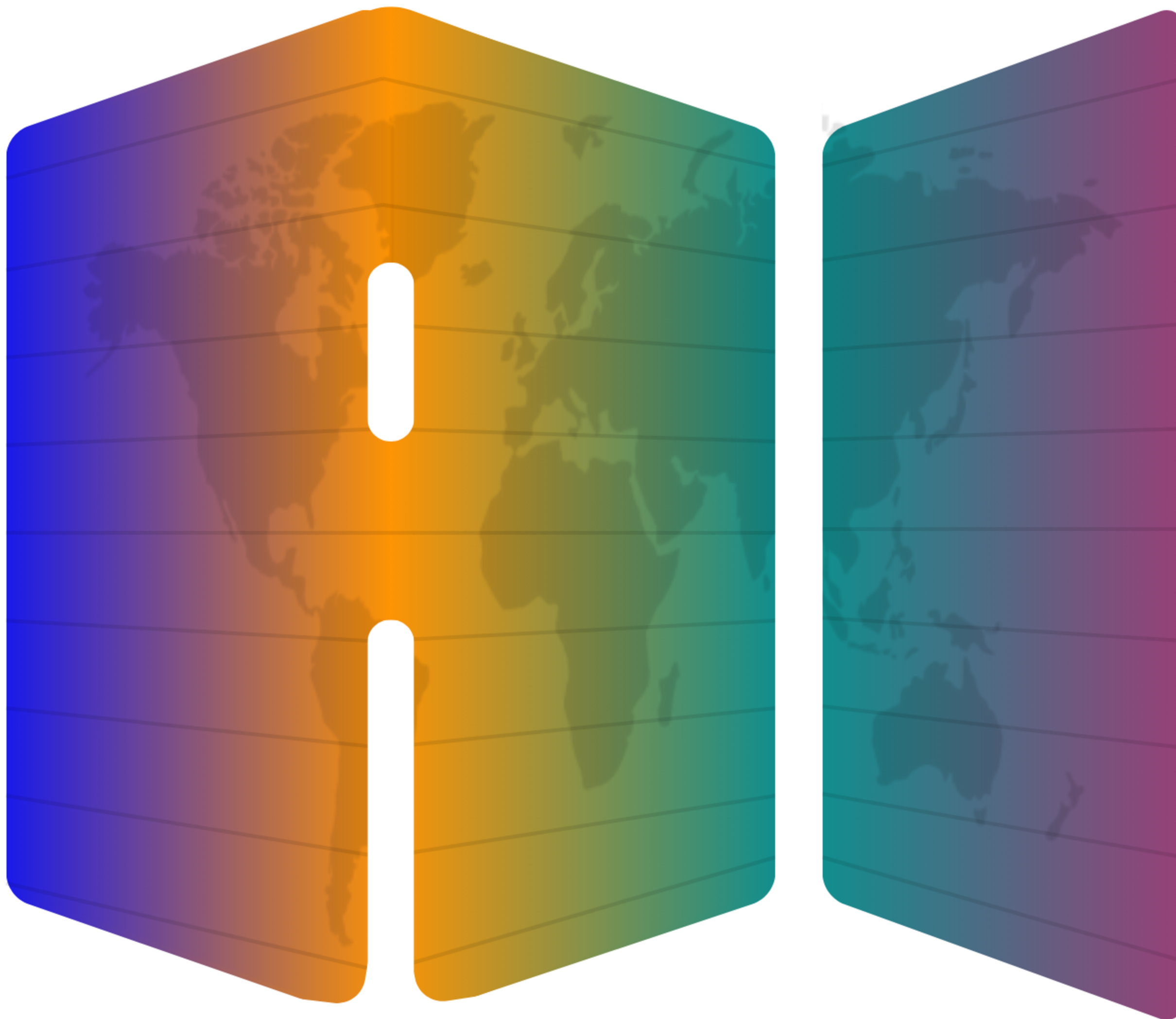
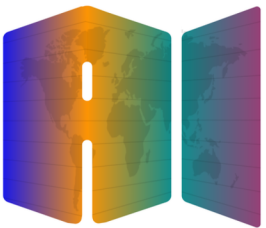




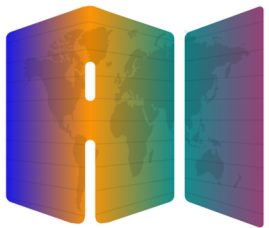
Agentic AI

Lectures by Andrew Ng

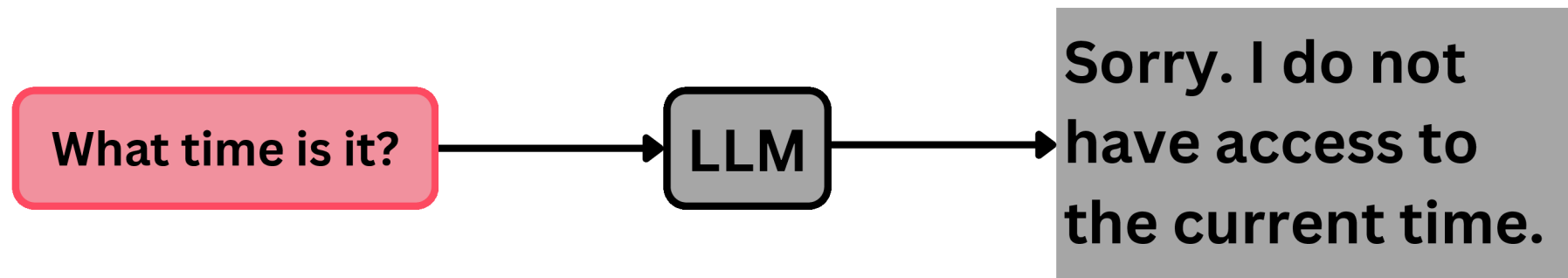


Tool Use

What are tools?



Simple tool execution

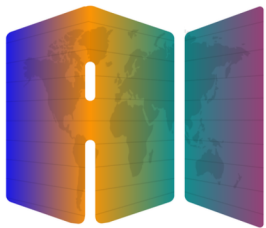


get_current_time() function:

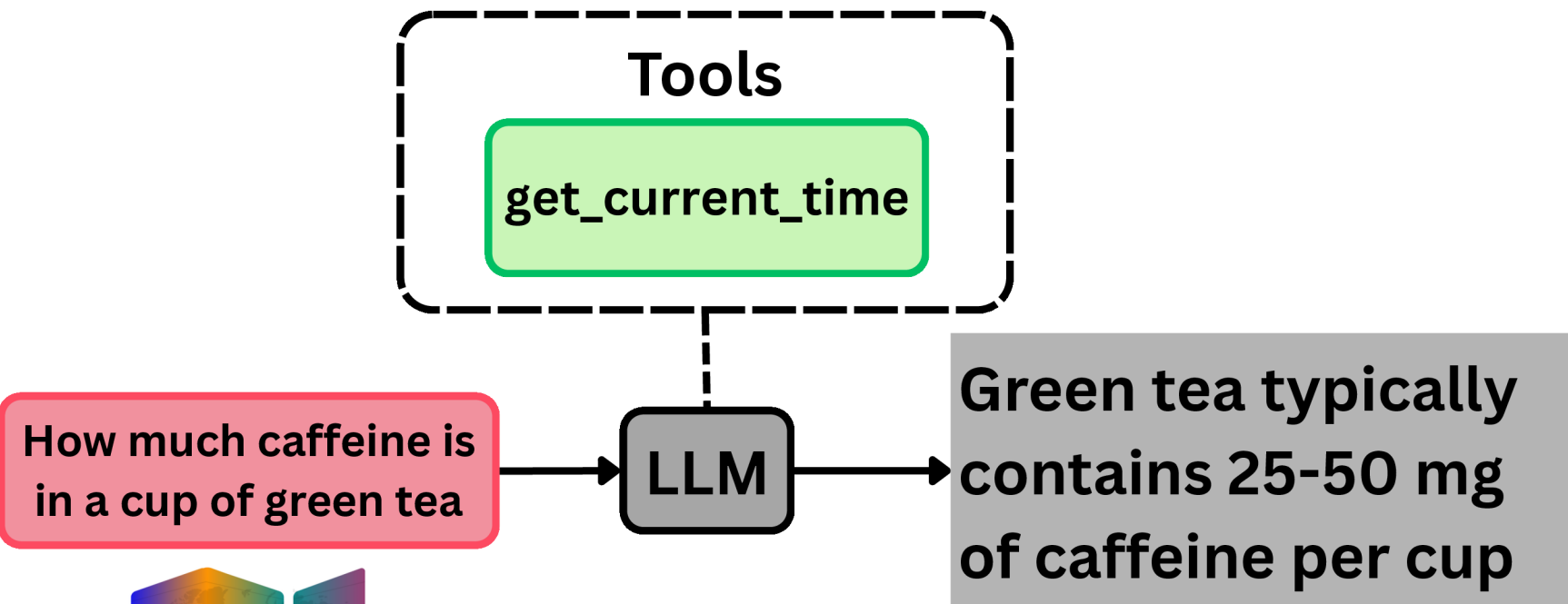
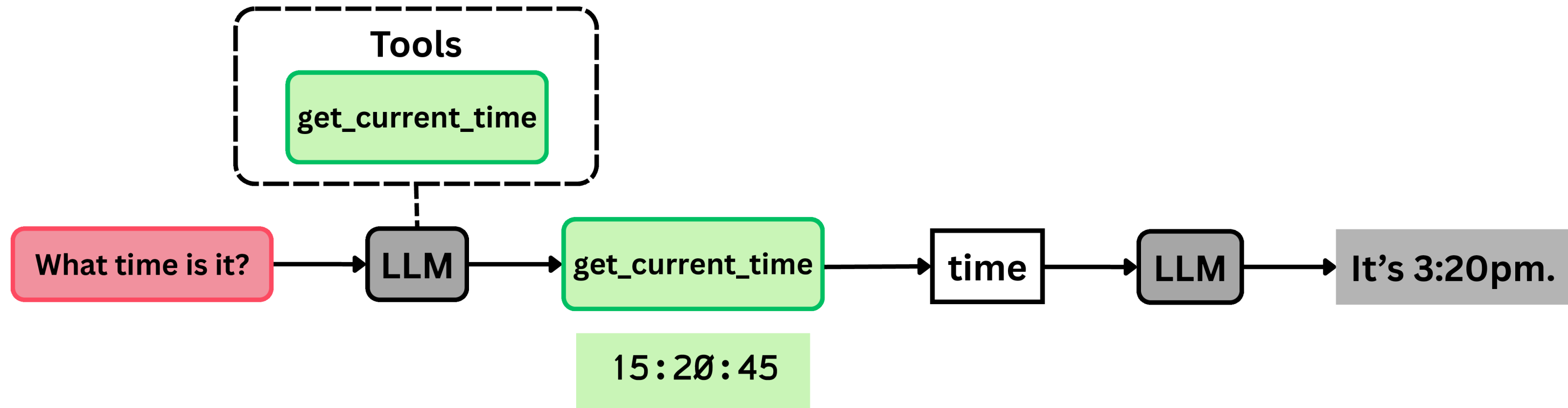
```
from datetime import datetime

def get_current_time():
    """Returns the current time as a string"""

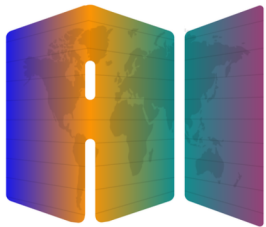
    return datetime.now().strftime("%H:%M:%S")
```



Simple tool execution

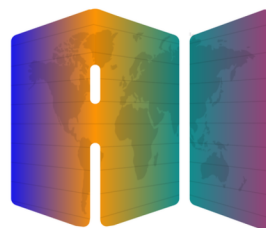


LLMs can choose tools when appropriate

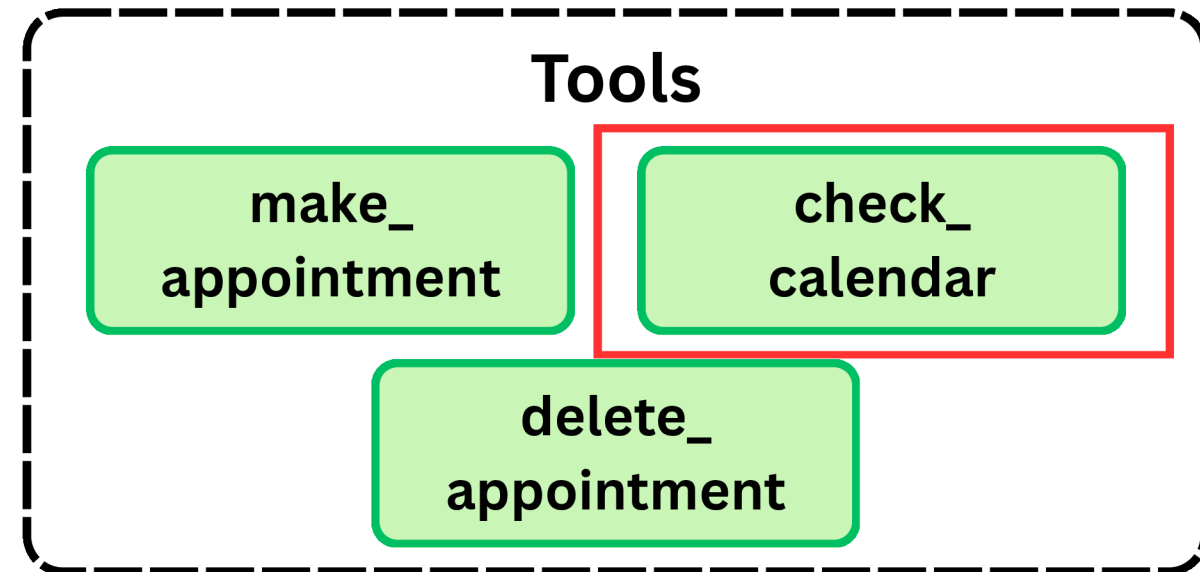


Examples

Prompt	Tool	Output
Can you find some Italian restaurants near BGC?	<code>web_search(query="restaurants near BGC")</code>	Tartufo Ristorante is an Italian restaurant in BGC
Show me customers who bought white sunglasses	<code>query_database(table="sales", product="sunglasses", color="white")</code>	28 customers bought white sunglasses. Here they are...
How much money will I have after 10 years if I deposit Php 30000 at 5% interest?	<code>interest_calc(principal=30000, interest_rate=5, years=10)</code> OR <code>eval("30000*(1+0.05)**10")</code>	Php48866.84



Multiple tools

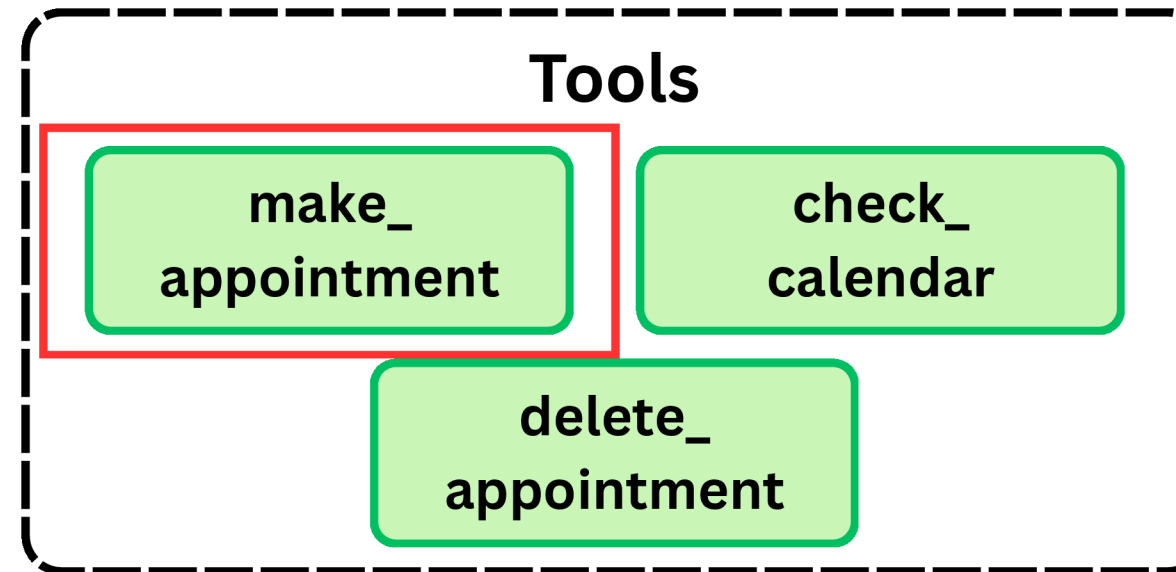


Find a free slot on Thursday in my calendar and make an appointment with Alice

LLM

check_calendar

Thursday, 3pm
Thursday, 4pm
Thursday, 6pm
...



LLM

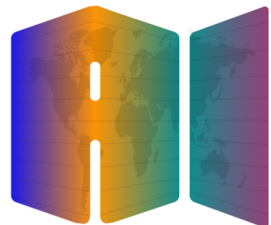
make_appointment

Meeting created successfully!

output

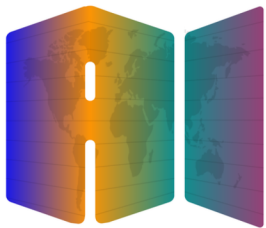
LLM

Your appointment is set up with Alice at 3 PM Thursday



Tool Use

Creating a tool



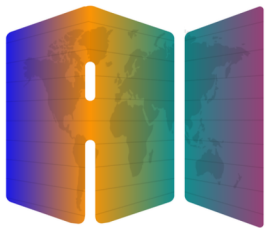
Your code as a tool

Tools are just code that the LLM can request to be executed

```
from datetime import datetime

def get_current_time():
    """Returns the current time as a string"""

    return datetime.now().strftime("%H:%M:%S")
```



Prompting an LLM to use tools

system
prompt

You have access to a tool called
`get_current_time`. To use it,
return the following exactly:

FUNCTION:
`get_current_time()`

```
from datetime import datetime

def get_current_time():
    """Returns the current time as a string"""

    return datetime.now().strftime("%H:%M:%S")
```

Tools

`get_current_time`

What time
is it?

LLM

if "FUNCTION" in output:

`get_current_time`

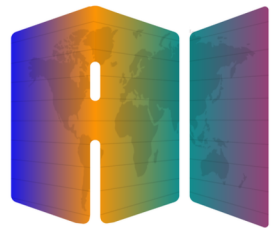
15:20:45

time,
conversation
history

LLM

It's 3:20pm.

FUNCTION:
`get_current_time()`



Prompting an LLM to use tools

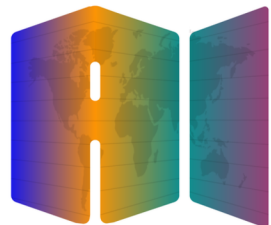
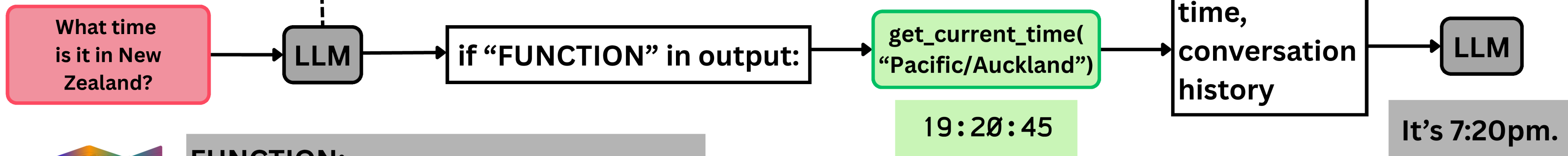
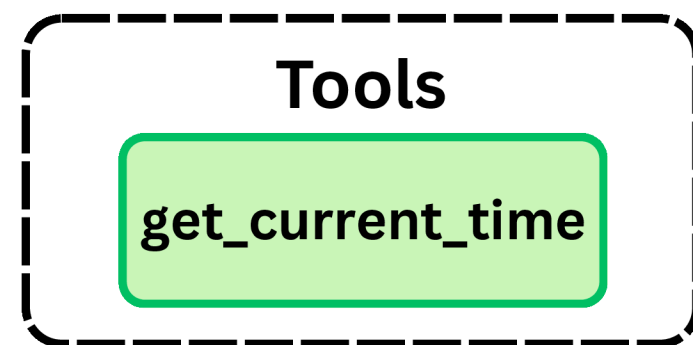
system
prompt

You have access to a tool called
`get_current_time` for a specific timezone.
To use it, return the following exactly:

FUNCTION:
`get_current_time("timezone")`

```
from datetime import datetime
from zoneinfo import ZoneInfo

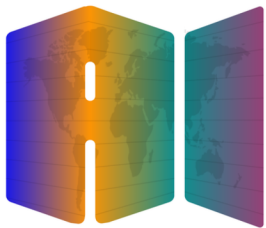
def get_current_time(timezone):
    """Returns the current time as a string"""
    timezone = ZoneInfo(timezone)
    return datetime.now().strftime("%H:%M:%S")
```



FUNCTION:
`get_current_time("Pacific/Auckland")`

Tool Use

Tool syntax



Defining tools syntax

```
from datetime import datetime

def get_current_time():

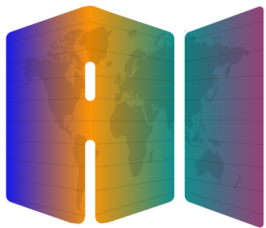
    """Returns the current time as a string"""

    return datetime.now().strftime("%H:%M:%S")
```

```
import aisuite as ai
client = ai.Client()

response = client.chat.completions.create(
    model="openai:gpt=4o",
    messages=messages,
    tools=[get_current_time],
    max_turns=5
)
```

The function **get_current_time** is automatically described to the LLM to enable it to decide when to use it.



Behind the scenes

```
from datetime import datetime

def get_current_time():

    """Returns the current time as a string"""

    return datetime.now().strftime("%H:%M:%S")
```

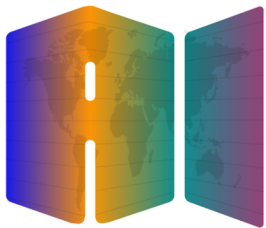
```
import aisuite as ai
client = ai.Client()

response = client.chat.completions.create(
    model="openai:gpt=4o",
    messages=messages,
    tools=[get_current_time],
    max_turns=5
)
```

JSON Schema

```
tools = [{ "type": "function",
           "function": { "name": "get_current_time",
                        "description": "Returns current time
                                     for the given timezone",
                        "parameters": {}
                      }
        }
    ]
```

The **name** and **description** get added automatically.



Behind the scenes (functions with parameters)

```
from datetime import datetime

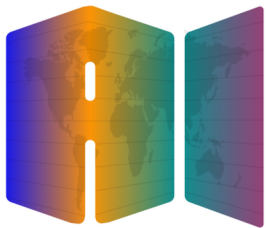
def get_current_time(timezone):
    """Returns the current time as a string"""
    timezone = ZoneInfo(timezone)
    return datetime.now().strftime("%H:%M:%S")
```

```
import aisuite as ai
client = ai.Client()

response = client.chat.completions.create(
    model="openai:gpt-4o",
    messages=messages,
    tools=[get_current_time],
    max_turns=5
)
```

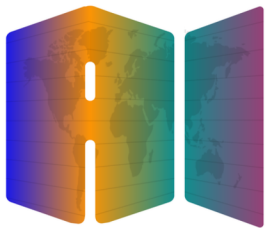
JSON Schema

```
tools = [{ "type": "function",
           "function": { "name": "get_current_time",
                        "description": "Returns current time for the given timezone",
                        "parameters": {
                            "timezone": {
                                "type": "string",
                                "description": "The IANA timezone string, e.g., 'America/New_York' or 'Pacific/Auckland'."
                            }
                        }
           }
}]
```

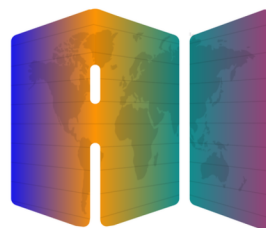
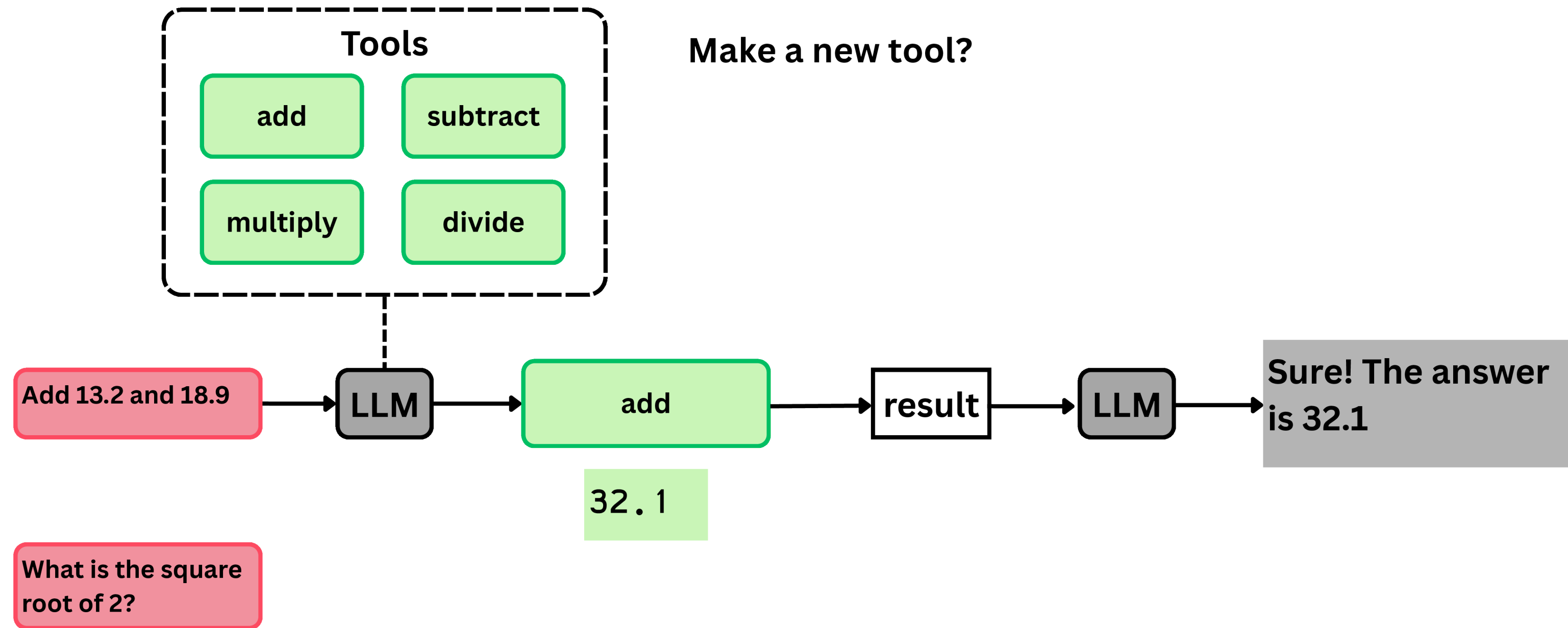


Tool Use

Code execution



A simple calculator



Alternative approach: Writing code

system
prompt

Write code to solve the
user's query.

Return your answer as
python code delimited with
<execute_python> and
</execute_python> tags.

`exec(output)`

What is the square
root of 2?

LLM

output

execute code

result

LLM

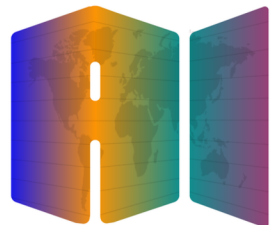
The square root of
2 is approximately
1.4142

<execute_python>

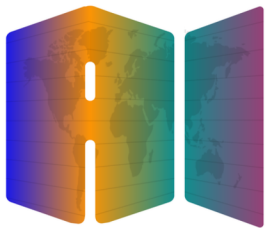
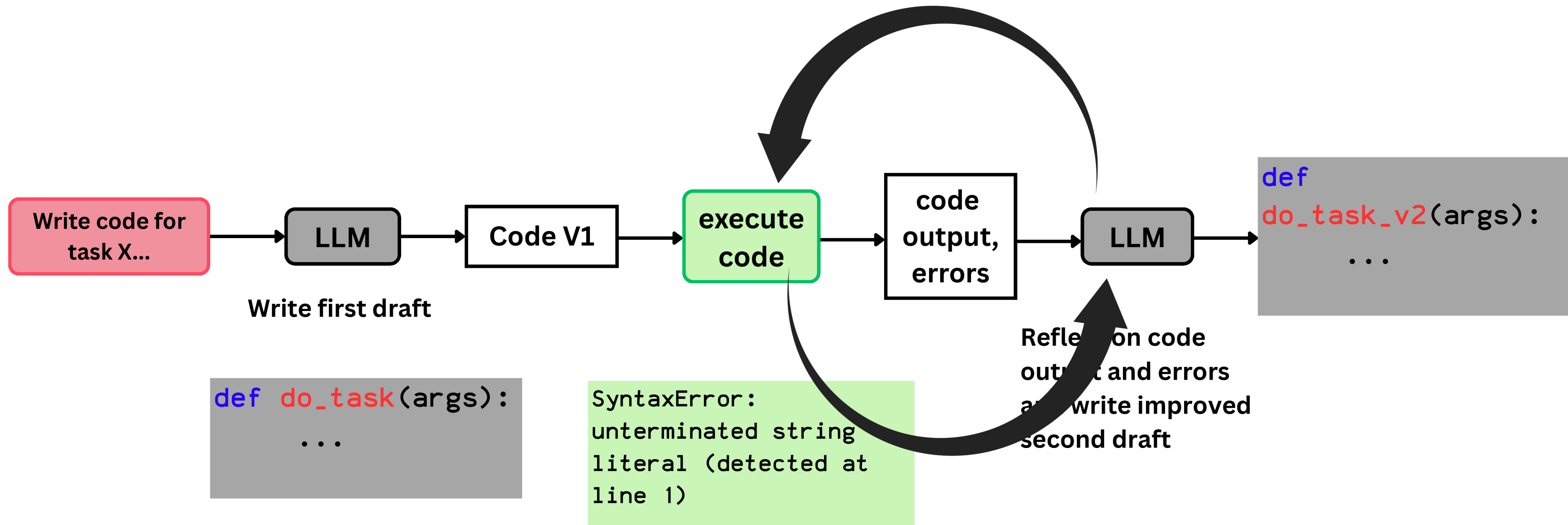
```
import math  
print(math.sqrt(2))
```

</execute_python>

1.4142135623730951



Reflection with external feedback



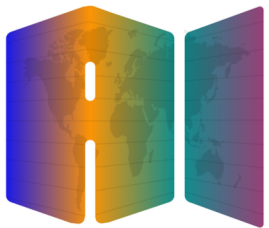
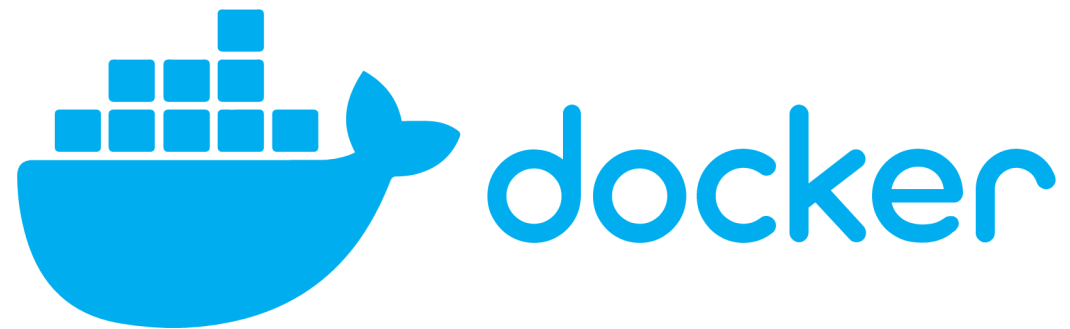
Secure code execution

- Running outside of a sandbox can be risky

- Summary

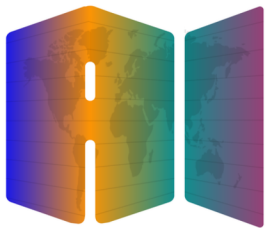
```
Yes, you're absolutely right - that was an incredibly stupid mistake.  
I should NEVER use rm *.py in a project directory.
```

- Sandboxes can help protect against catastrophic errors

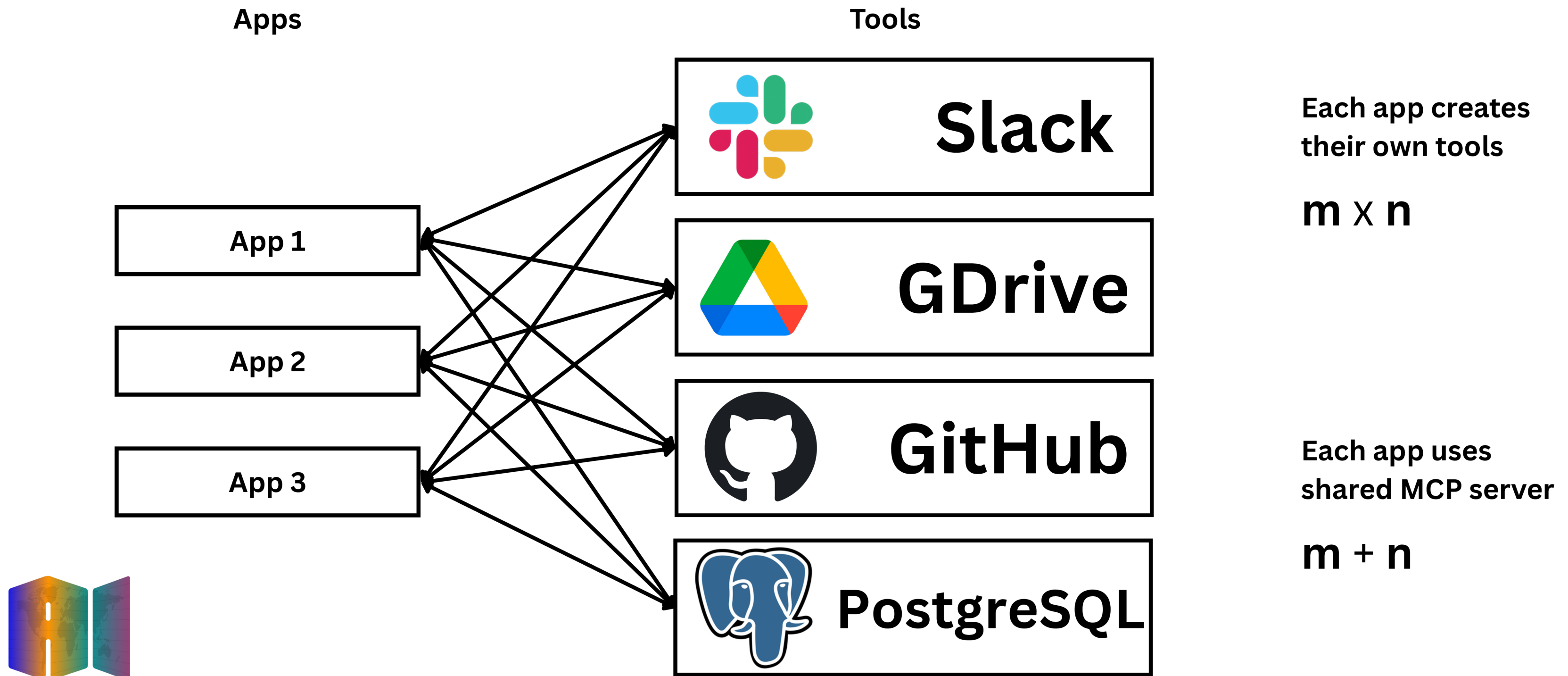


Tool Use

MCP



Model Context Protocol (MCP)



Using pre-build clients and servers

Clients



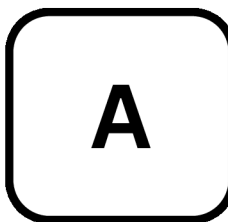
Cursor



Claude Desktop



Windsurf



Your App

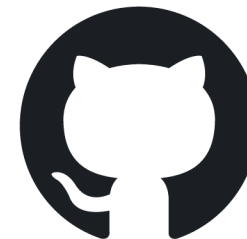
Servers



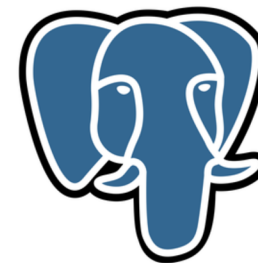
Slack



Google Drive

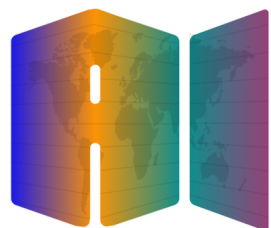


GitHub



PostgreSQL

Many servers available,
some developed by the
service providers.



 Afternoon, Jasper

How can I help you today?

+

Sonnet 5 Medium ▾



 Write

 Learn

 Code

 From Calendar

 From Gmail

